

CROWSNest, AB, Canada

WMO: 712360

Lat: 49.630N

Lon: 114.480W

Elev: 4275

StdP: 12.56

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-21.1	-13.4	-28.6	1.3	-20.4	-20.5	2.1	-12.4	27.6	28.4	25.5	28.3	2.9	130	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.8	83.6	59.3	80.5	58.5	77.2	57.5	62.9	76.8	61.0	75.2	59.2	73.4	9.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	84.8	66.9	55.8	77.7	65.2	53.7	71.9	63.4	30.6	76.9	29.1	75.2	27.8	73.7	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
22.7	20.0	18.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-31.1	89.3	6.9	3.0	-36.1	91.5	-40.2	93.2	-44.1	94.9	-49.1	97.1		
			-31.4	66.4	6.9	1.9	-36.3	67.8	-40.3	68.9	-44.2	69.9	-49.2	71.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	21.5	22.0	29.1	37.4	45.8	52.6	59.6	57.7	49.9	39.5	29.5	20.9
	DBStd	16.86	14.77	13.35	11.89	7.80	6.72	5.32	5.02	5.01	7.04	8.63	12.14	13.51
	HDD50	4841	883	784	647	381	165	32	3	5	87	334	616	904
	HDD65	9544	1348	1204	1112	829	595	372	177	230	453	790	1065	1369
	CDD50	784	0	0	0	2	36	110	299	244	83	9	1	0
	CDD65	12	0	0	0	0	0	0	9	3	0	0	0	0
	CDH74	1438	0	0	0	1	32	103	652	516	130	4	0	0
	CDH80	314	0	0	0	0	3	20	159	112	20	0	0	0
Wind	WSAvg	7.2	9.3	8.1	7.9	6.9	6.3	6.3	6.2	5.4	5.7	7.3	8.6	8.8
Precipitation	PrecAvg	27.4	2.0	1.7	2.0	2.0	3.1	3.8	1.9	1.8	2.0	2.0	2.5	2.1
	PrecMax	36.4	3.1	2.8	3.9	3.7	6.0	10.2	6.6	5.3	6.0	3.5	5.5	4.6
	PrecMin	20.1	0.6	0.5	0.7	1.1	0.6	1.5	0.3	0.4	0.2	0.8	0.6	0.9
	PrecStd	5.0	0.7	0.7	0.8	0.6	1.5	2.0	1.5	1.1	1.3	0.9	1.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	49.1	58.0	69.0	78.1	83.1	88.5	87.4	83.0	69.9	57.8	46.6
		MCWB	40.8	38.1	42.5	49.0	53.9	60.7	62.1	60.1	57.2	52.7	45.3	40.1
	2%	DB	43.8	44.4	51.9	63.5	73.0	76.5	84.3	83.1	77.8	63.4	52.0	42.6
		MCWB	36.9	36.4	40.4	46.4	52.3	57.5	60.3	58.7	56.0	49.1	43.2	36.8
	5%	DB	41.0	41.6	48.1	57.6	68.2	72.7	80.9	80.0	73.4	58.7	47.9	40.0
		MCWB	34.9	35.0	38.8	43.2	50.7	55.0	59.7	57.8	54.7	46.6	41.1	34.9
	10%	DB	38.8	38.7	44.5	52.5	63.5	68.4	77.4	76.4	68.0	54.3	44.4	37.2
		MCWB	33.6	32.7	36.9	41.2	48.7	53.3	58.5	57.2	52.5	44.6	38.4	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	40.0	44.8	50.2	56.5	63.1	66.5	64.0	60.4	54.1	47.8	41.4
		MCDB	48.0	45.5	54.0	65.6	72.5	77.1	80.1	78.2	75.8	64.8	54.0	45.8
	2%	WB	37.8	37.5	41.7	47.0	54.2	59.6	63.8	61.7	57.5	50.4	44.2	37.8
		MCDB	42.4	43.4	49.8	60.9	69.1	73.5	77.5	76.3	73.0	61.0	51.3	41.5
	5%	WB	35.9	35.3	39.6	44.4	51.9	56.9	61.9	60.0	55.4	47.6	41.5	35.1
		MCDB	40.0	41.1	46.8	55.7	65.4	68.7	75.6	74.8	71.0	57.0	47.2	39.2
	10%	WB	33.8	33.0	37.4	42.0	49.6	54.7	60.1	58.4	53.2	45.0	38.8	32.7
		MCDB	38.3	38.2	43.9	51.5	61.4	65.2	73.8	73.2	66.9	53.6	43.9	37.0
Mean Daily Temperature Range	5% DB	MDBR	18.1	20.3	20.2	22.2	24.8	24.5	29.8	31.3	27.5	21.5	16.4	16.5
		MCDBR	16.2	20.0	24.5	34.3	38.5	36.4	39.3	41.6	41.7	30.9	17.8	13.5
		MCWBR	11.7	13.7	14.7	18.6	19.2	18.2	17.9	19.2	20.9	17.1	11.4	10.1
	5% WB	MCDBR	14.3	19.4	22.8	31.1	33.9	31.7	33.9	35.6	38.2	28.8	17.1	13.0
		MCWBR	11.0	13.9	14.5	17.5	17.4	16.9	17.1	17.8	19.8	16.7	11.6	10.1
Clear-Sky Solar Irradiance	taub	0.239	0.250	0.278	0.331	0.339	0.331	0.330	0.341	0.308	0.281	0.259	0.231	
	taud	2.386	2.363	2.376	2.289	2.395	2.462	2.477	2.444	2.517	2.574	2.504	2.415	
	Ebn at Noon	267	290	298	290	292	294	292	284	285	277	258	255	
	Edh at Noon	20	26	31	38	36	34	33	32	27	21	18	16	
All-Sky Solar Radiation	RadAvg	349	657	989	1424	1691	1824	2130	1752	1218	729	368	273	
	RadStd	33	52	89	114	142	148	168	128	128	79	36	24	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page